ANSI/IES/IUVA LM-93-22

Illuminating ENGINEERING SOCIETY

APPROVED METHOD: OPTICAL AND ELECTRICAL MEASUREMENTS OF FAR UV-C EXCIMER SOURCES

AN AMERICAN NATIONAL STANDARD

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should be directed to IES.

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1.0 Introduction and Scope

1.1 Introduction

The ultraviolet-C (UV-C) spectrum is typically defined as being from 100 nm to 280 nm. This spectrum can be further divided into other defined UV-C ranges, such as vacuum-UV (100 nm to 200 nm) and far UV-C (200 nm to 230 nm). While "far UV-C" is not officially defined, this is the meaning of the term as used in this document.

The use of far UV-C optical radiation sources in various disinfection applications is a subject of increasing interest. The main reason is that this region of the UV spectrum (200 nm to 230 nm) offers a high rate of pathogen reduction, with much lower photobiological risk than that of longer-wavelength UV devices, and without the problems associated with ozone generation at shorter UV wavelengths.

Far UV-C optical radiation is commonly produced by excimer sources (krypton bromide, KrBr*, and krypton chloride, KrCl*). (The asterisk in the symbol indicates that the molecule is an excimer; see **Section 3.4 excimer lamp**). However, other radiation sources, such as LEDs, are expected to become available in the future.

Excimer lamps work on the principle of a dielectric barrier discharge. A short pulse of electrical high-energy discharge is formed in a quartz vessel that is filled with a rare gas (e.g., krypton or xenon) and typically with an additional small amount of a halogen (chlorine, bromine, or iodine). Within the high-energy discharge, excimers are formed that will emit specific wavelengths during decay. It is important to note that although the main energy is emitted at the typical wavelength of the filling mixture (e.g., 222 nm with KrCl* and 207 nm with KrBr*), there is radiation at wavelengths outside of the far UV-C range (i.e., shorter than 200 nm and longer than 230 nm).

The discharge vessel does not contain inner electrodes, and energy is supplied through a capacitive discharge by applying electrodes to the outside of the vessel. The drivers of such excimer lamps are always specific to a particular type of lamp. Typically, the drivers supply very high voltage pulses (2 to 30 µs long), on the order of 2 to 10 kV, depending on lamp power and design.

1.2 Scope

This Approved Method considers the specific measurement challenges and characteristics of far UV-C optical radiation sources and does not focus on the measurement of energy efficacy but on application-relevant data such as electrical, irradiance, spectral distribution, and angular distribution of the optical radiation source, including the driver. The main reason for this different approach (compared to that used for other UV-C optical radiation sources, like UV-C LEDs and low-pressure mercury lamps) is that other reliable measurement methods (e.g., in a sphere) to measure total output power in the far UV-C range are not yet established.

This document describes the procedures to be followed and the precautions to be observed in performing uniform and reproducible measurements of the electrical and ultraviolet optical radiation characteristics of far UV-C excimer sources predominantly emitting at a peak wavelength within the far UV-C range (200 nm to 230 nm). The wavelength range for the purposes of this document is 200 nm to 300 nm.

This Approved Method does not address changes in optical radiation properties over time.

This approved method does not address the measurement for the production of ozone.

This standard does not address the safety concerns associated with making these measurements. It is the responsibility of the user of this standard to establish appropriate safety and health practices. Additional information on safety with respect to UV radiation may be found in *ANSI/IES RP-27.1-22, Recommended Practice: Risk Group Classification and Minimization of Photobiological Hazards From Ultraviolet Lamps and Lamp Systems*.[1]

2.0 Normative References

2.1 ANSI/IES LM-75-19

Approved Method: IES Guide to Goniometer Measurements, *Types, and Photometric Coordinate Systems*. New York: Illuminating Engineering Society; 2019.

For measurements using a goniometer system, the laboratory shall meet the requirements stated in ANSI/IES LM-75-19.

2.2 ANSI/IES LM-91-22

Approved Method: Application Distance Radiometry. New York: Illuminating Engineering Society; 2022.

For measurements in a plane, the laboratory shall meet the requirements stated in ANSI/IES LM-91-22.

2.3 ANSI/IES LS-1-22

Recommended Practice: Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2021. Online: https://www.ies.org/standards/definitions/. (Accessed 2022 Feb 11).

2.4 CIE S 017/E:2020 ILV

International Lighting Vocabulary (ILV). Geneva: International Commission on Illumination (CIE); 2020.

3.0 Definitions

The definitions in the subsections that follow are specific to this document. Standard nomenclature and definitions, radiometric and photometric quantities, and illuminating engineering terminology are found in *ANSI/IES LS-1-22, Lighting Science: Nomenclature and Definitions for Illuminating Engineering* (see **Section 2.3**).

3.1 acceptance interval

The range of permissible measured quantity values. (See **Annex A** in this document, and Section 3.3.9 in ISO/IEC Guide 98-4.[2])

The acceptable results of a measurement lie within an acceptance interval, defined as the tolerance interval (see **Section 3.9**) reduced by the expanded uncertainty (95% confidence) of the measurement on both limits of the tolerance interval.

3.2 device under test (DUT)

The excimer lighting product being tested. It is typically an excimer source, including the driver recommended by the manufacturer (see **Figure 3-1**).

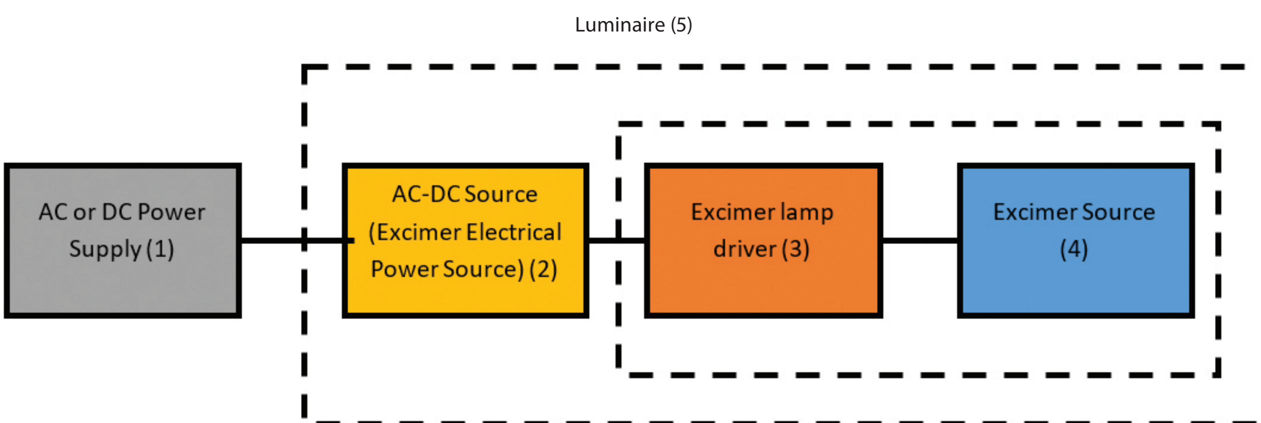

Figure notes:

1 - Laboratory electrical power source to regulate the electrical power into the DUT when testing a luminaire or similar end product.

2 – AC to DC (or DC to DC) electrical power source, contained within a luminaire or similar end product.

3 - Driver customized for the excimer source; typically combined with an excimer source.

4 - Excimer source, which can be a lamp or similar component.

5 – Far_UV-C luminaire.

Additional note: For some applications, the device could be a DC-input luminaire relying on a building's DC electrical power source instead of AC.

Figure 3-1. Schematic of UV-luminaire components.

Note: A lamp or luminaire would be considered an integrated device packaged as a single product.

In general, a far UV-C source should be considered as a combination of a lamp (or module) and a specific driver for such lamp. For the purposes of this document, a lamp is a far UV-C source that will only be used inside a fixture or luminaire.

3.3 excimer lamp

A source of ultraviolet radiation, produced by dimeric or heterodimeric molecules that have been excited by a high-voltage discharge (thus becoming excimer[†] molecules) and then produce UV radiation as they transition to their ground state. The wavelength of UV radiation emitted depends on the type of noble gas, which is, depending on the type of source, combined with a halogen gas. Krypton-chloride excimer (chemical symbol: $KrCl^*$) lamps emit primarily at 222 nm, with lesser emissions at longer UV wavelengths. (As noted in **Section 1.1 Introduction**, the asterisk in the symbol indicates that the molecule is an excimer.)

3.4 excimer power source

A device that provides the current that powers the DUT during measurements.

3.5 lamp

A generic term for a manufactured source created to produce optical radiation.

Note: For the purposes of this document, unless stated otherwise, the term refers specifically to excimer lamps (see **Section 3.3 excimer lamp**).

3.6 luminaire, UV

For the purpose of this document: A lamp system that distributes, filters, or transforms the UV radiant energy transmitted from at least one source of optical radiation and which includes all the parts necessary for attaching and protecting the sources and, where necessary, circuit auxiliaries together with the means for connecting them to the power supply.

[†] An excimer is an extremely short-lived molecule consisting of two atoms or molecules linked together, at least one of which is a noble gas or other type of species with a valence shell completely filled with electrons.

3.7 reference surface of DUT

The outer optical surface of the DUT. For example, this will be the surface of a lamp, or the surface of a quartz window or filter of a UV luminaire.

Note: This is not necessarily the reference plane aligned with the inverse-square law (refer to **Section 8.2**).

3.8 solar blind detector

In the context of this document: a photodetector that is insensitive to UV-A, visible, and infrared radiation but responds specifically to ultraviolet radiation with wavelengths shorter than 300 nm.

3.9 tolerance interval

The interval of permissible values of a property. (See **Annex A** in this document, and ISO/IEC Guide 98-4,[2] Section 3.3.5.)

Note 1: In this document, stated conditions include a tolerance interval.

Note 2: The term tolerance interval as used in conformity assessment, such as in this document, has a different meaning from the same term as used in statistics.

Physical and Environmental Test Conditions

4.1 General

Excimer lamps are not very sensitive to temperature or operating position. However, since the output of the lamp will also depend on driver temperature, it is recommended that lamps be measured only after thermal equilibrium of driver and lamp has been achieved by measurement of lamp output for stability as defined in **Section 8.3 DUT Stabilization**.

4.2 Ambient Temperature

The ambient temperature in the test environment shall be measured at a point not more than 1 m from the lamp and driver and at the same height as the lamp and driver. Under standard conditions, this temperature shall be maintained at 25 °C ± 5 °C. In addition, the temperature sensor shall be shielded from the lamp and from radiation from any other source. Measurements at temperatures other than 25 °C ± 5 °C are considered nonstandard and shall be documented in the test report.

4.3 Airflow

Air movement at the surface of a lamp and driver under test may alter electrical and radiant power values. No discernible airflow from sources other than the device under test shall be allowed unless it is specifically defined by the manufacturer of the optical radiation source. This can be tested using a single-ply tissue hanging in place of the lamp (with the lamp removed). To ensure stabilized and uniform air convection over the lamp surface, the lamp should be operated in the manufacturer specified orientation when measurements are being taken.

For goniometer measurements that require movement of the device under test, the instantaneous tangential velocity of any point on the DUT shall be less than 0.20 m/s.

When active cooling is used based on the manufacturer's specification, the cooling method shall be reported.

4.4 Vibration

No specific requirements are stated, but good laboratory practice requires that DUTs not be subjected to excessive vibration or shock during stabilization, transportation, mounting, or testing.

4.5 Stray Optical Radiation (Including Stray Light)

For all measurements, stray optical radiation should be suppressed in the test environment. Since surface reflectance assumptions in the far UV-C range are unreliable, only shielding and baffling should be considered. In addition, stray optical radiation may be measured and subtracted from the DUT measurement. (Refer to ANSI/IES LM-75-19 for more detailed information and requirements; see **Section 2.1**.)

Note 1: Due to the unknown UV-C reflectance performance of black paint or black fabric, such material or finishes should not be relied on to reduce stray optical radiation. This is subject to change as more information is obtained on such finishes and/or materials through testing.

Note 2: Ambient optical radiation can affect the measurements of far UV-C products. Steps should be taken to limit the ambient optical radiation in the test environment.

Note 3: UV-C lamps can stimulate fluorescence if susceptible (i.e., inappropriate) materials are used or through contamination.

4.6 Humidity and Barometric Pressure

Relative humidity values greater than approximately 65% in the defined ambient temperature range (see **Section 4.2**) can lead to corrosion effects in some instruments, and values below approximately 10% can lead to electrostatic effects. Therefore, laboratory humidity should be monitored and maintained between 10% and 65%.

Since the ignition and operation of excimer lamps depends on very high voltages, very low air pressure can influence the measurements. It is recommended to monitor and report the air pressure. In general, measurements above 2,000 m in elevation should be avoided unless specifically desired.

Electrical Test Equipment

5.1 Electrical Power Supply Requirements

5.1.1 Voltage Waveshape and Frequency. The input voltage to the DUT shall follow the manufacturer's specification. If AC voltage is required, the AC power supply shall have a sinusoidal voltage waveform at the prescribed frequency (typically 60 Hz or 50 Hz) such that the total harmonic distortion shall not exceed 3% of the fundamental frequency during operation of the DUT. The supplied frequency shall have a tolerance interval of ±2 Hz from the prescribed frequency.

If the input current to the DUT is DC, the AC ripple of the DC voltage in a frequency range of 10 Hz to 1 kHz shall be <1%.

5.1.2 Voltage Regulation. The voltage of an electrical power supply (DC or RMS voltage) applied to the DUT shall be regulated to within ±0.2% under load. If AC electrical power is used, the AC electrical power supply shall have a current crest-factor capability greater than required by the DUT. If the current crest factor of the waveform required by the DUT is unknown, the electrical power supply shall have a current crest-factor capability of at least 5.

If the input voltage to the DUT is DC, the DC voltage at the input to the DUT shall be within ±0.2% of the specified voltage. It is recommended that a DC electrical power supply that can provide a current of at least 2 times the rated current of the DUT be used.

5.2 Test and Reference Circuit Requirements for DUT

The test setup can affect the measurement results. This setup will vary depending on whether the DUT is a luminaire or is a combination of a driver and an excimer source. *The instructions in this section are for a DUT that is the latter: an excimer source combined with a specific driver.* The serial number or designation of the excimer source and driver shall be documented.

The total electrical input power of the DUT shall be measured. As shown in **Figure 5-1**, for AC input power an AC power analyzer capable of measuring electrical power, current, and voltage and an electrical power supply shall be connected to the input terminal of the DUT. The wiring of the test circuit should position the DUT within the manufacturer specified distance. The length of the wires shall be recorded. Since the DUT is a combination of a driver and an excimer lamp, it is recommended to avoid running wires in parallel over significant distances from the driver to the lamp and from the lamp to the ground plane within close vicinity of each other. Conducting wires that are close in proximity may cause a capacitance that can create variances in the electrical power delivered to the lamp. If using a grounded electrical power supply system, the measurement circuit shall be grounded where indicated by the manufacturer. The lengths of the wires between driver and excimer lamp shall not exceed the manufacturer's recommendations.

Note: The calculation of electrical power by multiplying current and voltage is not acceptable for AC supply due to the power factor in such a system not being equal to 1.0. For DC supplies, this method gives sufficient accuracy.

The output of excimer lamp drivers is a very high-voltage, pulsed source; the current is also a pulsed, very high-frequency low current; and drivers may have very specific grounding requirements. Therefore, the use of standard electrical power analyzers to measure the excimer lamp electrical input is very often not reliable and for this reason not recommended. The electrical input power should be measured as the input to the excimer lamp driver.

All electrical measurement equipment shall be calibrated and traceable to the International System of Units (SI).

5.2.1 Voltage Circuit Internal Impedance. To avoid error due to leakage currents, the internal impedance of voltage measurement circuits (which includes the electrical power analyzer) shall be at least 1 MΩ as

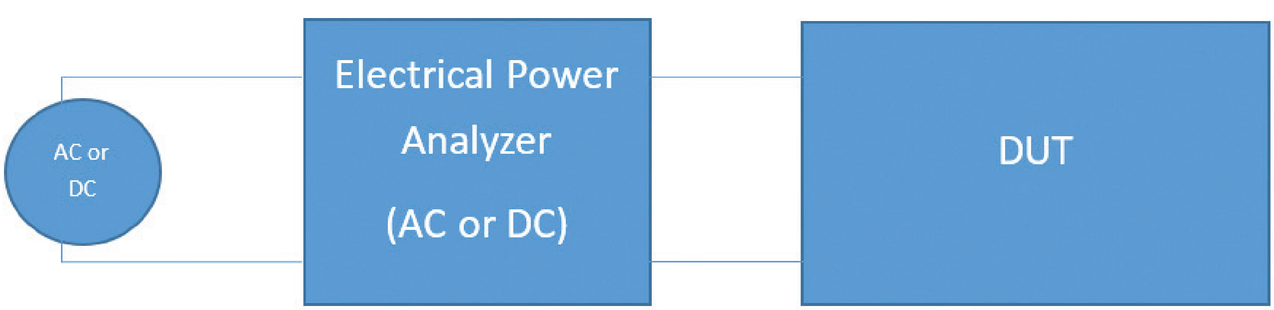

Figure 5-1. Measurement circuit for DUT.

measured by disconnecting the electrical power supply and measuring the resistance at the driver electrical input.

5.2.2 AC Electrical-Power Analyzer Uncertainty. AC electrical-power analyzers shall be operated with all line filters and frequency filters disabled.

For the measurement of RMS AC voltage, the analyzer shall have an expanded uncertainty (k = 2) of 0.4% or less for measurement of a 60-Hz sinusoidal waveform.

Note: Most AC electrical-power analyzers on the market provide specifications in terms of accuracy. A discussion of the relationship between accuracy and measurement uncertainty is provided in **Annex A – Tolerance Interval vs. Acceptance Interval**.

6.0 Optical Measurement and Equipment

Since far UV-C sources should not be assumed to be radiators of narrow optical bandwidth, it is necessary to measure the spectral output of the DUT and report it. The spectral output can then be used for further detailed calculations according to guidelines from *ANSI/IES RP-27.1-22, Recommended Practice: Risk Group Classification and Minimization of Photobiological Hazards From Ultraviolet Lamps and Lamp Systems,*[1] *IEC 62471-6, Photobiological Safety of Ultraviolet Lamp Products,*[3] or the American Conference of Governmental Industrial Hygienists (ACGIH).

For performing optical measurements, either a spectroradiometer or radiometer may be used.

The spectroradiometer or radiometer shall have a cosine-corrected angular responsivity. Cosine response can be achieved by using a light-transmitting diffuser as the first optical element. The cosine response of the detector should have a directional response error of less than 2% within the angular subtense of the lamp.

6.1 Spectroradiometer

6.1.1 Minimum Requirements. The spectroradiometer shall fulfill the following minimum requirements:

a. Spectral range between 200 nm to 300 nm
b. Spectral data point interval ≤ 1 nm
c. Bandpass ≤ 3.0 nm
d. Wavelength uncertainty < 0.5 nm ($k = 2$)

In addition:

e. The spectroradiometer should account for light outside of this wavelength range that may result in stray light within the spectroradiometer system, especially during calibration. Internal stray optical radiation correction may be required for array-spectroradiometers.
f. The linearity of the array spectroradiometer should be tested (refer to Section 4.2.6 in ANSI/IES LM-58-20[4]).

6.1.2 Wavelength Calibration. The spectroradiometer shall be calibrated for wavelength following the manufacturer's procedures with accepted wavelength calibration sources. The bandpass of the spectroradiometer should be set to the smallest possible value before calibration and remain unchanged for the measurement of DUTs. The wavelength calibration at 222 nm should be verified by measuring and recording the spectral irradiance of a KrCl* lamp, which is 222.0 nm ± 0.2 nm. The calibration should also be checked and recorded at 253.7 nm using a low-pressure mercury lamp to confirm the instruments wavelength scale.

Note: Ultraviolet bandpass filters do not attenuate at a constant value across the UV spectrum and will cause a spectral shift in the emission peak. The KrCl* lamp must have the bandpass filter removed to check the emission peak at 222 nm.

6.1.3 Spectral Irradiance Calibration. The spectroradiometer shall be calibrated for spectral irradiance responsivity traceable to the International System of Units (SI).

The spectroradiometer shall be calibrated with a suitable calibration source (e.g., a deuterium lamp) in the same optical configuration as will be used for the optical measurements. The calibration shall follow the manufacturer's procedures. If optical fibers are used, a cosine adapter at the end of the fiber shall be used. The integrity of the fiber should be verified, and a mechanically secure connection to the spectroradiometer should be ensured. The calibration shall be verified by re-measuring the spectral irradiance calibration source (see **Section 6.1.2**) after calibration.

A spectroradiometer that uses an optical fiber may suffer from solarization. Therefore, the wavelength of calibration should be considered based on the degree of solarization.

Special care should be taken to minimize the effect of excessive noise and drift during the calibration by optimizing integration time and averaging multiple scans. The bandpass of the spectroradiometer shall be set to the same values as will be used during the optical measurements.

6.2 Radiometer

The radiometer is a spectrally integrating detector that may have a spectral filter to limit the wavelength range of detection (e.g., to avoid detection of ambient optical radiation or undesired out-of-band optical radiation). A solar blind radiometer would be ideal.

The radiometer shall fulfill the following minimum requirements:

a. Spectral sensitivity between 200 nm and 300 nm (solar blind detector)

b. Signal-to-noise ratio > 1,000:1

In addition:

c. Radiometer saturation should be evaluated

d. The linearity of the radiometer should be tested

If measurements are being performed at short distances from the DUT, the linearity and maximum measurement range of the detector should be considered.

6.2.1 Calibration. The radiometer shall be calibrated for irradiance responsivity traceable to the International System of Units (SI) through an NMI. (Refer to CIE 220:2016, *Characterization and Calibration Methods of UV Radiometers*.[5]) The calibration method shall take into consideration the geometry and spectral power distribution of the calibration source as compared to the DUT's primary wavelength. A spectral mismatch correction shall be applied to minimize error.

6.2.2 Detector Linearity. The radiometer should be used within the linear range per manufacturer specifications. If used outside the manufacturer specified range, the linearity shall be verified. (Refer to *CIE 237:2020, Non-Linearity of Optical Detector Systems* for guidance on testing for linearity.[6])

Note: The radiometer is composed of a detecting component and an amplifier that converts the signal into a detectable quantity. Both parts can experience non-linearity effects.

6.2.3 Detector Spectral Response. Since excimer lamps should not be assumed to be monochromatic and may have significant spectral irradiation between 200 nm and 270 nm, the spectral response of the irradiance detector can be a significant error source and shall be considered. Very narrow-band filtered detectors (e.g., 222 nm ± 5 nm) are not recommended, since the uncertainty of the filter spectral response is high and will filter substantial parts of the 222-nm peak.

Note: Measurements of undesirable optical radiation require a radiometer with a filter that blocks wavelengths shorter than the undesired wavelengths. However, for determining quantities that are wavelength dependent (e.g., threshold limit values) a spectroradiometer is preferred.

6.2.4 Sensitivity of the Detector to High Electromagnetic Interference (EMI). Excimer lamps typically operate at high frequency (30 kHz to 100 kHz) with very high voltages. This may create high EMI in a frequency range of 30 kHz to 1 Mhz. If the detector is placed very close to an excimer lamp, the measurement results may be altered by such interference. Steps should be taken to reduce the level of interference caused by the DUT on the detector. In such scenario, the EMI sensitivity should be evaluated by inserting a non-transmissive part (e.g., > 12 mm polycarbonate or acrylic) between the DUT and the detector and the response of the detector recorded. Scattered optical radiation shall be eliminated to avoid a false EMI signal.

7.0 Test Preparation

7.1 DUT Orientation During Seasoning

Lamp seasoning and optical measurements shall be done in the same orientation as in the intended measurement method and shall follow the manufacturer's recommendations when provided. The DUT orientation shall be reported.

7.2 DUT Seasoning

Seasoning for new DUTs shall follow the manufacturer's recommendations. If no recommendations are available, a minimum of 2 hours of seasoning should be utilized. The DUT seasoning duration shall be reported. For situations where a DUT is being tested with unknown operation or seasoning time (for example, for a DUT returned from the field for evaluation), the report shall state that the total operation or seasoning time, respectively, is unknown.

8.0 Measurement Methods

8.1 Mounting Orientation for Measurement

The DUT shall be tested in the operating position with respect to gravity recommended by the manufacturer or for the intended use of the excimer product. Stabilization and optical measurements of the DUT shall be performed in the same operating position. The position and orientation with respect to a goniometric system of the DUT as mounted for measurement shall be reported.

The UV luminaire shall be mounted in a way that prevents supports or other adjacent objects from obstructing the detector's access to direct optical radiation or reflecting any other optical radiation into the detector. Where a moving-mirror goniometer is used, the detector shall be shielded so, that that radiation from the luminaire's optical assembly can reach the detector only by reflecting from the mirror. (See Figure 6-7 in ANSI/IES LM-75-19) This will help to minimize extraneous optical radiation reaching the detector.

Note: Some products can be challenging to measure in the orientation of intended use. In such instances where an alternate orientation is necessary, deviations shall be reported.

8.2 Determination of the Optical Axis and Reference Plane

The luminaire shall be mounted on the goniometer so that the following referenced plane or parts of the luminaire coincide with the center of the goniometer:

- Case 1, recessed and coffered luminaires: The opening in the major plane, see **Figure 8-1 (left).**
- Case 2, direct luminaire: The plane of the major optical radiation-emitting opening, see **Figure 8-1 (right).**

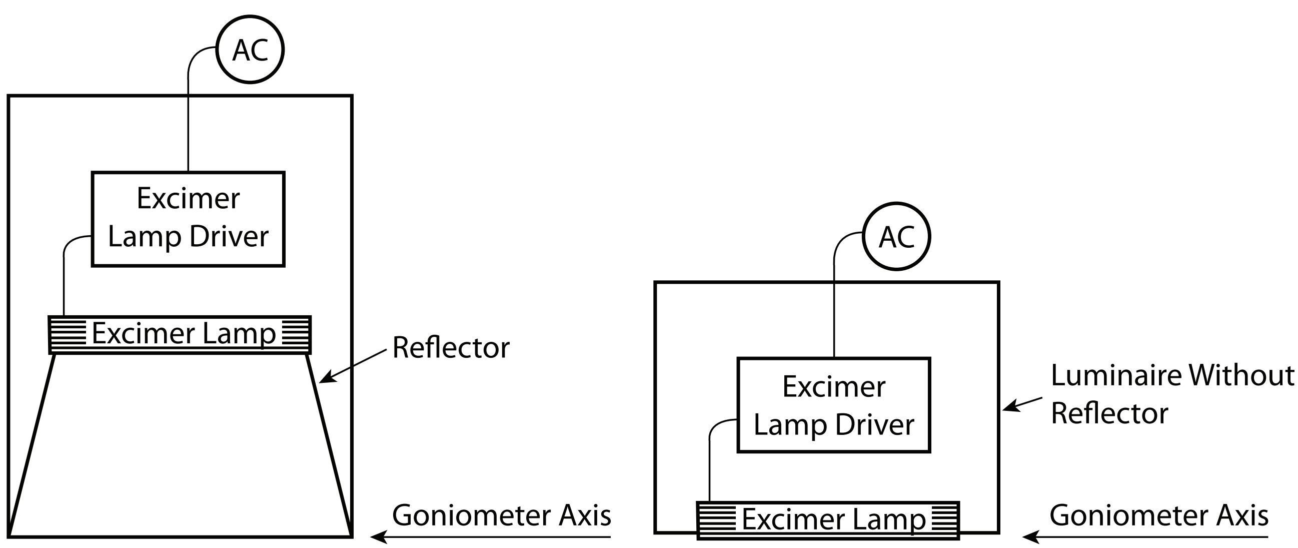

Figure 8-1. Positioning of a luminaire on the goniometer for a recessed lamp (left) and for a surface-mount direct lamp (right).

8.3 DUT Stabilization

The DUT shall be stabilized prior to taking a measurement. Stability shall be achieved when the variation (maximum to minimum) of at least three readings of the optical radiation and electrical power consumption, taken at a maximum of 1-minute intervals over a period of 3 minutes and divided by the last of these measurements chronologically, is less than 1%. Readings should be taken at regular intervals. The stabilization time shall be reported. Any deviations shall be reported.

8.4 UV Irradiance Measurement Distance

Measurements shall be made in a far-field condition (where a single emitter follows the inverse-square law). The standard measurement distance shall be 1.0 m from the reference surface of the detector to the reference plane of the DUT (see **Section 8.2 Determination of the Optical Axis and Reference Plane**).

For a DUT with angular distributions that are nearly Lambertian, a test distance greater than five times the longest optical-radiation emitting dimension should be sufficient. Longer test distances may be required for beam forming DUTs. Since the optical scheme used to produce beam forming DUTs can be complex, it is recommended that the minimum required measurement distance be determined by experimentally measuring the variation of intensity with distance to find the minimum distance at which the inverse-square law applies. The test distance shall be reported.

Excimer UV luminaires can vary in construction, with some having a single excimer emitter and others with more than one. **Figure 8-2** shows a sketch of a DUT with two emitters. The longest optical-radiation emitting dimension is the distance between the two emitters; therefore, the minimum measurement distance should five times this distance. Alternatively, the emitters may be operated independently, and the test report shall show the results of each independent measurement, with the measurement device centered perpendicularly to each emitter.

Measurements in a near-field condition are covered in ANSI/IES LM-91-22 (see **Section 2.2**).

8.5 UV Irradiance Measurements

The DUT shall be mounted as identified in **Sections 8.1, 8.2,** and **8.4,** and stabilized as identified in **Section 8.3**. The test setup should minimize the effect of stray optical radiation as identified in **Section 4.5**. The UV irradiance measurements should be measured using a spectroradiometer (see **Section 6.1**) or a radiometer (see **Section 6.2**). At a minimum, a data point shall be taken perpendicular to the DUT at 1.0 m.

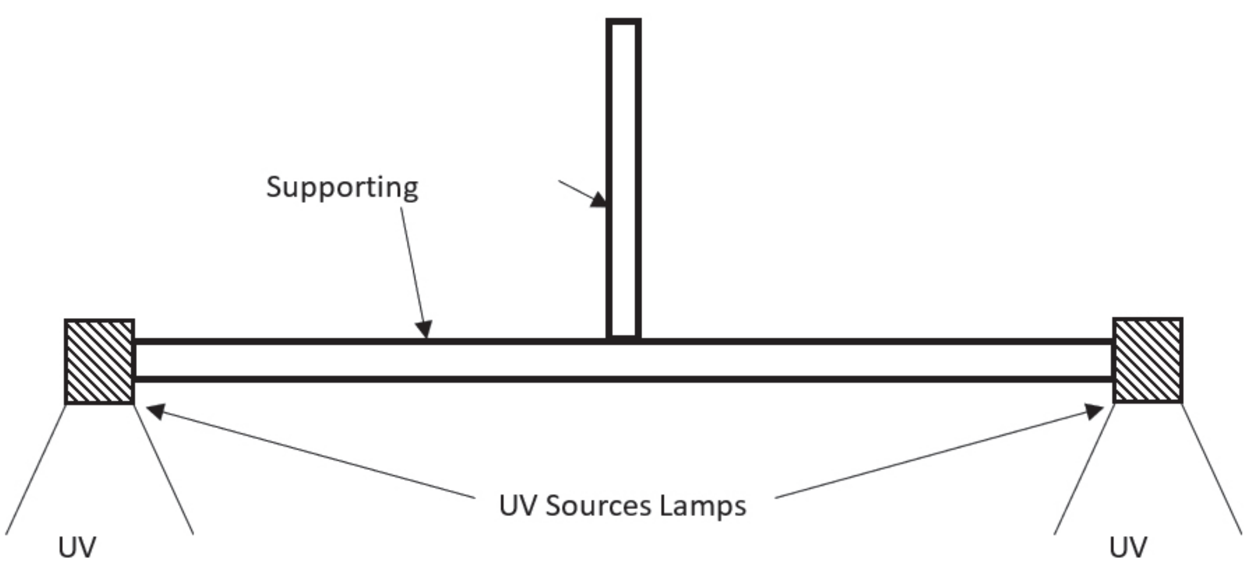

Figure 8.2. Schematic of a DUT with two emitters.

8.5.1 Goniometer Measurements. The goniometer measurements are the preferred method for generating IES files where the measurements follow the inverse-square law (see ANSI/IES LM-63-19[7] or ANSI/IES TM-33-22[8]).

The goniometer type shall be capable of maintaining the intended operating position unchanged with respect to gravity; therefore, only Type C goniometers shall be allowed (refer to ANSI/IES LM-75-19; see **Section 2.2**). The distance requirement is not critical if only integrated quantities are to be measured.

Care should be exercised to prevent optical radiation reflected from the mechanical structure of the goniometer or any other surface, including secondary reflections from surfaces of the DUT itself, from reaching the detector. The speed of rotation of the positioning equipment shall be such as will minimize the disturbance of the thermal equilibrium of the DUT (see **Section 4.3**).

The optical center of the DUT shall be aligned with the intersection of the goniometer axes. A description of the location of the optical center of the DUT and a description of the orientation of the DUT with respect to the goniometer axes shall be reported. The goniometer should have sufficient angular resolution and absolute alignment to characterize the DUT. The required angular resolution and absolute alignment of the goniometer system is dependent on the slope of the DUT's radiant intensity distribution with respect to angle.

The goniometer axis (the rotation axis of the DUT holder) and the position of the detector should be accurately co-aligned. This should be checked periodically, as the goniometer axis can drift over time if the mirror angle deflects. Even a small-angle deflection of the mirror can cause large errors in the measurement of a narrow-beam DUT.

Mathematical modeling is used in applications to stitch together data from several emitters in a DUT that is too large to be measured directly. *Reports shall include measured data only.*

Note: A Type D goniometer is acceptable only for prototyping and characterizing an excimer DUT. Typical excimer devices have shown a 2% change in optical radiation as the DUT is rotated with respect to gravity. *This nominal change is too large for rating purposes.*

8.5.1.1 Angles for Measurement. The vertical increments around the DUT shall be sufficiently small that the data taken describe the optical radiation distribution adequately. Vertical steps of 5 degrees are advised, but smaller increments (2.5° or smaller) shall be used if necessary. Vertical steps of 1° are advised whenever the beam angle is less than 30°. In no case shall vertical steps exceed 10°.

8.5.1.2 Planes for Measurement. The number of vertical planes explored shall be determined by the degree of asymmetry or irregularity existing within the optical radiation distribution being measured. Where a DUT is asymmetric, a series of vertical half-planes shall be used to describe the distribution accurately. At a minimum, readings should be taken in 16 vertical half-planes spaced 22.5° apart. Highly asymmetric distributions may require a larger number of half-planes. In all cases, as many half-planes as are necessary to describe the optical radiation distribution adequately shall be taken. For DUTs that have two planes of symmetry (are quadrilaterally symmetric), five planes are usually adequate (measured horizontally with respect to the DUT axis at 0° [parallel], 22.5°, 45°, 67.5°, and 90° [normal]). For bilaterally symmetric distributions (with one plane of symmetry), it will be necessary to report measurement planes from 0 through 180 degrees, with 22.5-degree steps usually adequate. The DUT position should be clearly indicated with respect to the beam position for the DUT, such as for wall washers. For electronic published data, it is recommended that the 0-degree position be on the beam side.

8.5.2 Planar Measurements. For a defined distance from the reference plane of the DUT, a plane is defined with the reference plane of the detector positioned normal to the DUT. Within this plane a measurement grid shall be defined. (Refer to ANSI/IES LM-91-22 for direction, particularly Section 7.4.1; see **Section 2.2** of this document for the normative reference.)

In the cases of high-power devices measured at small distances or for long test durations, it is recommended that multiple dark measurements be taken.

For certain purposes, such as simple graph generation, data may be collected with Cartesian coordinates – i.e., a rectangular scan. Once the measurement grid is defined, the irradiance shall be measured at each location on the grid, according to LM-91-22. Any data values less than 0 shall be set to 0 for display purposes.

If it is assumed that the emission of the DUT will demonstrate 1-fold or 2-fold mirror symmetry, then a half-plane or quarter-plane, respectively, may be captured and mirrored for reporting purposes.

(To determine the total optical radiation flux on the measured area, use Section 8.0 of ANSI/IES LM-91-22)

Note: The accuracy of measurements for broadband sources may be affected by the spectral sensitivity of the radiometer. In such cases, the accuracy may be improved by establishing a correction factor for the radiometer's spectral measurements. ANSI/IES RP-27-20[9] (Sections 6.2.2 and 6.2.2.1) or IEC 62471-6 (Annex C.6) provide more guidance on determining such a correction factor.

9.0 Measurement Uncertainty

If a statement of uncertainty is required, the recommendations given in ISO/IEC Guide 98-3:2008 should be followed.[10] For all measurements, expanded uncertainty with a confidence interval of 95% shall be used—thus, in most cases, using the coverage factor $k = 2$.

To verify the measurement uncertainty, the measurement results of some DUTs may be compared with measurements traceable to an NMI.

The distance uncertainty shall be less than 5.0 mm or less than 0.5%, whichever is smaller.

10.0 Test Report

The report shall list all pertinent data concerning conditions of testing. The report shall also list the type of equipment, the DUT, and the reference standards used.

Typical items reported are:

a. Date of measurement and testing agency
b. Manufacturer's name and designation of the DUT
c. Measured quantities: e.g., total radiant flux, radiant efficiency, peak wavelength, centroid wavelength
d. Rated electrical values: electrical input power, voltage, current, and power factor
e. Instrument used
f. Designation and type of reference standard used
g. Correction factors applied (e.g., spectral mismatch)
h. Optical measurement conditions (e.g., goniometer test distance)
i. Measurement parameters: input current, forward voltage, and electrical power
j. Spectral power distribution, if applicable
k. Bandwidth of the spectroradiometer, if spectral distribution is reported
l. Statement of uncertainties, if required
m. Deviation from standard operating procedures, if any
n. Stabilization method used
o. DUT electrical power source (ballast or driver)
p. Far UV-C DUT type (e.g., filtered, non-filtered)

For special purposes, it may be desirable to determine the characteristics of lamps when they are operated at other than the standard conditions described in this Approved Method. Where this is done, such results are meaningful only for the conditions under which they were obtained, and these conditions shall be stated in the test report.

Informative Annex A – Tolerance Interval vs. Acceptance Interval

This annex is not part of ANSI/IES LM-93-22. It is included for informational purposes only.

In this document, required conditions are stated in terms of a tolerance interval, with specified upper and lower tolerance limits. To ensure that any given parameter is within the specified tolerance interval, the applicable measurement uncertainty shall be considered by deriving the corresponding acceptance interval. The acceptance interval is defined as the tolerance interval reduced by the expanded uncertainty of measurement (at 95% confidence) at both limits of the tolerance interval. This relationship is illustrated graphically in **Figure A-1**. The measured value of any specified parameter shall lie within the acceptance interval derived from the corresponding tolerance interval and measurement uncertainty.

As an example, the specifications for a commonly available AC electrical power analyzer provide the accuracy of the AC voltage measurement for frequencies ranging from 45 Hz to 66 Hz as:

$$Accuracy = (0.1\%)\ (Reading) + (0.1\%)\ (Range)$$

This accuracy is valid for three months. For a 12-month calibration accuracy, the 3-month accuracy is multiplied by 1.5. This accuracy is revalidated when a calibration laboratory performs a single-point calibration or verification of a range setting on the AC electrical-power analyzer.

Therefore, for a 120-volt, 60-Hz reading (*Reading* = 120 V) on a 150-volt range (*Range* = 150 V), the accuracy is 0.41 V for 12 months, which defines the half-width of a uniform or rectangular distribution. Converting the half-width of a uniform distribution to standard uncertainty is done by dividing the half-width by the square-root of three. The standard uncertainty for this AC electrical-power analyzer for AC voltage measurement is 0.23 V, or 0.19%. With a coverage factor of $k = 2$, the expanded uncertainty or calibration uncertainty is 0.46 V, or 0.38%. For measuring 220 V, the expanded uncertainty is 0.90 V, or 0.41%.

To set the AC voltage to 120.00 V using this AC electrical-power analyzer, for which the tolerance interval is ±0.5% (as provided in **Section 5.2.2**), the AC electrical-power analyzer should read between 119.86 V and 120.14 V (which defines the acceptance interval). Further information on the concept of acceptance interval is provided in ISO/IEC Guide 98-4.[2]

Another example is the measurement of the ambient temperature, which according to **Section 4.2** shall be maintained at 25 °C with a tolerance interval of ±1.2 °C. If the expanded uncertainty ($k = 2$) of the thermometer is 0.2 °C, the reading of the thermometer shall be ± 1.0 °C from 25 °C, as shown in **Figure A-2**.

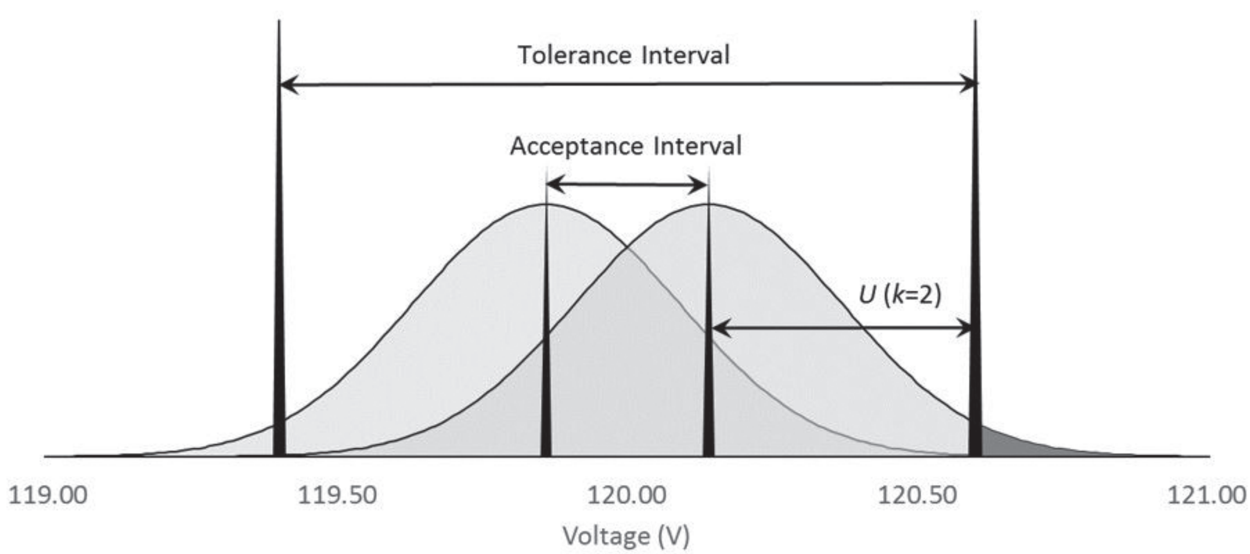

Figure A-1. Graphical relationship between tolerance interval, acceptance interval, and measurement uncertainty for setting 120 V AC.

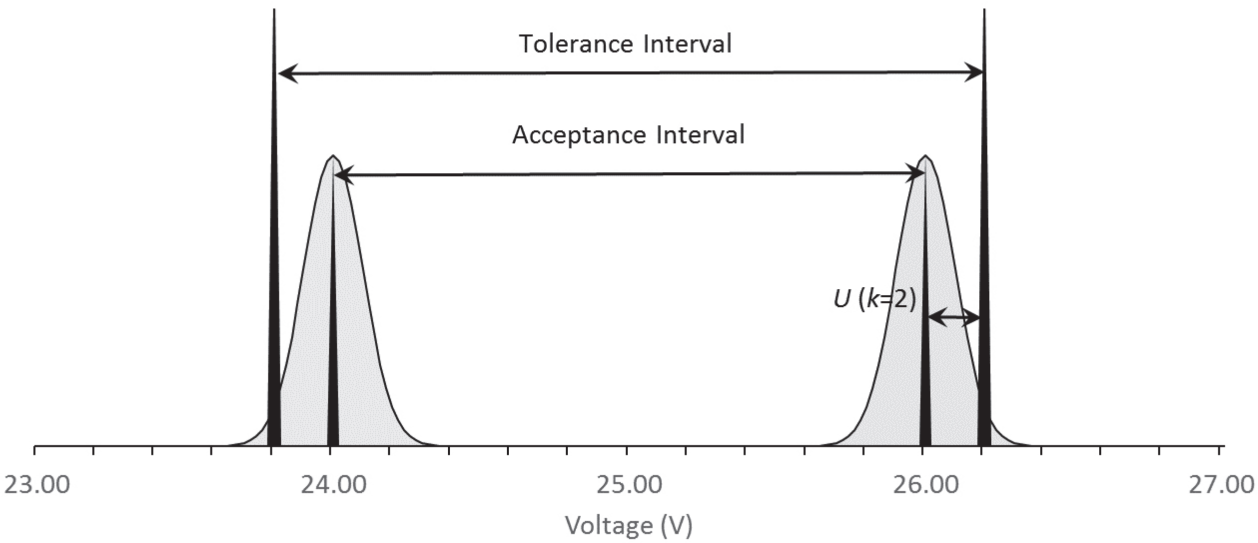

Figure A-2. Graphical relationship between tolerance interval, acceptance interval, and measurement uncertainty for ambient temperature measurement.

INFORMATIVE REFERENCES

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2. International Standards Organization (ISO) and International Electrotechnical Commission (IEC). Uncertainty of Measurement – Part 4: Role of Measurement Uncertainty in Conformity Assessment. Geneva: IEC; 2012. (ISO/IEC Guide 98-4: 2012. Also: JCGM 106:2012).

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5. International Commission on Illumination (CIE). CIE 220:2016, Characterization and Calibration Methods of UV Radiometers. Geneva: CIE; 2016.

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7. Illuminating Engineering Society. ANSI/IES LM-63-19, Approved Method: Standard File Format for the Electronic Transfer of Photometric Data and Related Information. New York: IES; 2019.

8. Illuminating Engineering Society. ANSI/IES TM-33-22, Technical Memorandum: Standard Format for the Electronic Transfer of Luminaire Optical Data. New York: IES; 2022.

9. Illuminating Engineering Society. ANSI/IES RP-27-20, Recommended Practice: Photobiological Safety for Lighting Systems. New York: IES; 2020.

10. International Standards Organization (ISO) and International Electrotechnical Commission (IEC). ISO/IEC Guide 98-3:2008. Uncertainty of Measurement — Part 3: Guide to the expression of uncertainty in measurement (GUM:1995). Geneva: International Standards Organization. (JCGM/WG1/100).

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

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2. Title of publications and year published ______________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________
4. My proposal (check one):
   [ ] Change to read as follows
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Use underscore to show material to be added (<u>added</u>) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____
____
____

6. Reason and substantiation:

____
____
____

Select as applicable:
   [ ] Additional pages are attached. Number of additional pages:
   [ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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